$$\mathsf{check}_j^k(\mathrm{tt}, v) \dashrightarrow v \qquad \text{Check-True}$$

$$\mathsf{check}_j^k(\mathrm{ff}, v) \dashrightarrow \mathsf{blame}_j^k \qquad \text{Check-False}$$

$$\mathsf{mon}_j^{k,l}(\mathsf{flat}(e), v) \dashrightarrow \mathsf{check}_j^k(e\ v, v) \qquad \text{Mon-Flat}$$

$$\mathsf{mon}_j^{k,l}(\kappa, v) \dashrightarrow \mathsf{guard}_j^{k,l}(\kappa, v) \qquad \text{Mon-Guard}$$

$$\text{where } \kappa \neq \mathsf{flat}(e)$$

$$\mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v_1)\ v_2 \dashrightarrow \mathsf{mon}_j^{k,l}\left(\kappa_2, v_1\ \mathsf{mon}_j^{l,k}(\kappa_1, v_2)\right) \qquad \text{Guard-Func}$$

$$\mathsf{guard}_j^{k,l}(\kappa_1 \to_i \lambda x.\kappa_2, v_1)\ v_2 \dashrightarrow \mathsf{mon}_j^{k,l}\left(\kappa_3, v_1\ \mathsf{mon}_j^{l,k}(\kappa_1, v_2)\right) \qquad \text{Guard-Dep}$$

$$\text{where } \kappa_3 = \kappa_2\left[x := \mathsf{mon}_j^{l,j}(\kappa_1, v_2)\right]$$

$$\frac{E \neq \square}{E\left[\mathsf{blame}_j^k\right] \twoheadrightarrow \mathsf{blame}_j^k} \text{ Step-C-Blame}$$

$$\frac{e_1 \dashrightarrow e_2}{E[e_1] \twoheadrightarrow E[e_2]} \text{ Step-C} \qquad \frac{e_1 \dashrightarrow e_2}{e_1 \dashrightarrow e_2} \text{ Step-C-Base}$$

Figure 1: Reduction Rules for Contracts